

Problem-Solution Fit

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem–Solution Fit Canvas

The **Problem–Solution Fit Canvas** is a strategic design tool used to validate whether the proposed solution effectively addresses the real problems experienced by the target users. It helps identify customer needs, challenges, behaviors, existing solutions, and the root causes of the problem before designing the final solution.

For the **Credit Card Approval Prediction using Machine Learning** project, the canvas focuses on customers applying for credit cards and financial institutions responsible for evaluating applications. The proposed machine learning system aims to simplify the approval process by providing quick, accurate, and data-driven predictions based on applicant information.

PROBLEM–SOLUTION FIT CANVAS				
1 CUSTOMER SEGMENT(S) [CS]  - Individuals applying for credit cards. - Bank credit officers evaluating applications. - Financial institutions issuing credit cards. - Organizations seeking automated credit risk assessment.	2 PROBLEMS / PAINS + ITS FREQUENCY [PR]  - Customers are uncertain about their eligibility before applying. - Manual verification takes significant time and effort. - Banks receive a large number of applications daily. - Human evaluation may lead to inconsistent decisions. - High frequency due to increasing demand for credit card services.	3 TRIGGERS TO ACT [TR]  - Need for faster credit card approval decisions. - Increasing number of customer applications. - Demand for automated decision-making. - Requirement to reduce manual workload and processing time.	4 EMOTIONS BEFORE / AFTER [EM]  Before - Uncertain - Worried - Confused - Impatient After - Confident - Satisfied - Informed - Relieved	5 AVAILABLE SOLUTIONS PROS & CONS [AS]  Existing Solutions - Manual credit verification - Traditional banking approval systems - Rule-based eligibility checking Pros - Well-established process - Human verification - Regulatory compliance Cons - Slow processing - Time-consuming - Higher operational cost - Possibility of human error
6 CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]  - Limited understanding of approval criteria. - Lack of financial knowledge. - Time constraints. - Limited access to instant eligibility assessment. - Dependence on manual banking procedures.	7 BEHAVIOR + ITS INTENSITY [BE]  - Customers compare different credit card offers. - Applicants repeatedly check eligibility. - Banks process numerous applications every day. - High demand for quick and reliable approval decisions. Intensity: HIGH	8 CHANNELS OF BEHAVIOR [CH]  ONLINE - Bank websites - Mobile banking applications - Credit card portals - Financial service websites OFFLINE - Bank branches - Customer service centers - Financial advisors - Banking agents	9 PROBLEM ROOT / CAUSE [RC]  - Manual evaluation consumes significant time. - Multiple applicant attributes make decision-making complex. - Large application volumes increase processing delays. - Difficulty in accurately identifying creditworthy applicants. - Lack of automated prediction systems.	
10 YOUR SOLUTION [SL]  The proposed Credit Card Approval Prediction using Machine Learning system uses a trained Random Forest Classifier to analyze applicant information and predict whether a credit card application is likely to be Approved or Rejected. The system is integrated with a Flask-based web application that provides a simple interface for users to enter their details and receive instant prediction results.  Accurate Prediction Uses ML model to deliver highly accurate approval predictions.  Fast Decisions Provides instant results and reduces manual processing time.  Easy to Use User-friendly web interface for seamless interaction.  Consistent & Reliable Eliminates human bias and ensures consistent decision-making.  Supports Better Decisions Helps customers and banks make informed and confident decisions.				